Summarization of Clinical Question–Answer Documents Using Extractive and Abstractive Transformer Models

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Abstract

Consumers seeking medical advice online frequently encounter long, complex, and redundant answers from multiple sources, making it difficult to quickly identify reliable information. Automatic summarization offers a promising solution to reduce cognitive load and present concise, evidence-based insights. In this work, we study four complementary summarization tasks derived from the MediQA 2021 shared task: (1) single-answer extractive, (2) single-answer abstractive, (3) multi-answer extractive, and (4) multi-answer abstractive summarization. We develop a hybrid summarization pipeline combining a BERT-based sentence classifier for extractive selection and a Flan-T5 model for abstractive rewriting. To address the challenge of extremely long clinical answers, we adopt a hierarchical strategy for multi-answer summarization in which single-answer summaries are fused into a final combined summary. We evaluate our approach using ROUGE and BERTScore. Results show that the hierarchical method improves multi-answer performance by mitigating input truncation and preserving key medical facts. Our findings highlight the benefits of lightweight transformer models for clinical consumer QA summarization in resource-constrained environments such as Google Colab.

1 Introduction

Patients commonly rely on online health resources such as MedlinePlus, Healthline, or Mayo Clinic for medical information. However, a single clinical question often yields multiple long answers written in different styles, with varying degrees of redundancy and reliability. Manually synthesizing these answers can be time-consuming and cognitively demanding.

Automatic summarization provides a mechanism to condense lengthy consumer-facing clinical content while maintaining correctness and clarity. Following the structure of the MediQA 2021 shared task, we focus on generating: (1) single-answer extractive summaries, (2) single-answer abstractive summaries, (3) multi-answer extractive summaries, and (4) multi-answer abstractive summaries.

We propose a hybrid architecture built from widely available transformer models that can be trained within Colab hardware limits. Our approach is designed to handle long inputs, noisy consumer text, and scenarios where input answers exceed standard transformer length limits.

2 Problem Definition

Given a question q and a set of answers $A = \{a_1, a_2, \ldots, a_n\}$, we define:

- **Single-answer extractive**: Select salient sentences within an answer.
- **Single-answer abstractive**: Rewrite the answer concisely in natural language.
- **Multi-answer extractive**: Extract sentences across all answers.
- **Multi-answer abstractive**: Produce a unified summary that fuses information across answers.

Challenges include extremely long clinical articles (up to 17k tokens), inconsistent writing style, and redundancy across answers.

3 Dataset

We use instructor-provided data derived from the MediQA 2021 shared task. Each question is associated with multiple answers, their sections, and gold extractive and abstractive summaries.
All data files are uploded on Github: **Link**

Each question in the dataset includes:

- A consumer health question
- Multiple reference answers curated from MedlinePlus and similar sources
- An abstractive multi-answer summary
- An extractive multi-answer summary

For each answer:

- Abstractive summary
- Extractive summary
- Section text
- Full article text
- Source URL
- Quality rating

3.1 Data Statistics

- Questions: 110 train, 16 validation, 30 test
- Answer-level training pairs: 392 train, 51 validation, 109 test
- Sentence-level extractive pairs: 12,429 train, 1,565 validation, 3,279 test

3.2 Length Analysis

Using the Flan-T5 tokenizer:

- Single-answer mean length: 843 tokens
- Single-answer truncated at 512: 45.9%
- Multi-answer mean: 2,914 tokens
- Multi-answer truncated at 512: 92.7%

These statistics highlight the need for hierarchical summarization and sentence-level extractive modeling.

4 Evaluation Metrics

We evaluate using:

4.1 ROUGE

ROUGE-1, ROUGE-2, and ROUGE-L:

$$ROUGE-N = \frac{overlappingN-grams}{referenceN-grams}$$

4.2 BERTScore

We compute contextual similarity using:

$$P = \frac{1}{|P|}\sum_i \max_j \cos(p_i, r_j), \quad R = \frac{1}{|R|}\sum_j \max_i \cos(r_j, p_i),$$

$$F1 = \frac{2PR}{P+R}$$

5 Models and Methods

5.1 Extractive Summarization

We fine-tune `longformer-base-4096` as a sentence-level classifier. Sentences are labeled positive if they match the gold extractive summary via fuzzy similarity. At inference time, we score all sentences and select the top-k most salient ones.

5.2 Abstractive Summarization

We fine-tune `google/flan-t5-base` using a question-conditioned input format. The model generates abstractive summaries up to 128 tokens. Due to Colab memory constraints, the maximum encoder length is set to 512.

5.3 Hierarchical Multi-Answer Summarization

Because full multi-answer concatenations exceed model length limits, we use:

1. Generate extractive and abstractive summaries for each answer.
2. Concatenate these shorter summaries.
3. Apply Flan-T5 again to generate a unified multi-answer summary.

This approach reduces input size by more than 80% and improves overall semantic coverage.

6 Results

6.1 Quantitative Results

Task	**R-1**	**R-2**	**R-L**	**BERT F1**
Single Extractive	0.52	0.43	0.47	0.90
Single Abstractive	0.49	0.34	0.41	0.89
Multi Extractive	0.38	0.23	0.27	0.86
Multi Abstractive	0.39	0.21	0.27	0.87

Table 1: ROUGE and BERTScore results across the four tasks.

7 Results Visualization

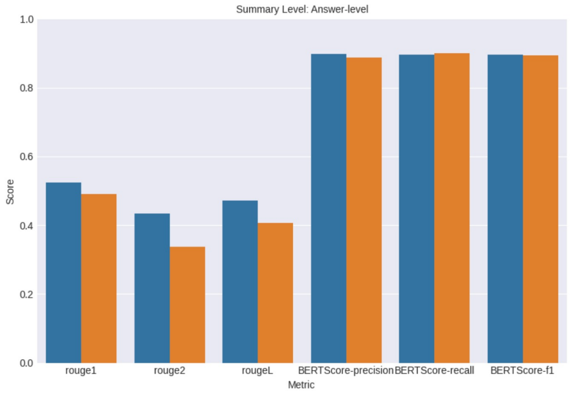

Figure 1: Metric comparison for Answer-level summarization.

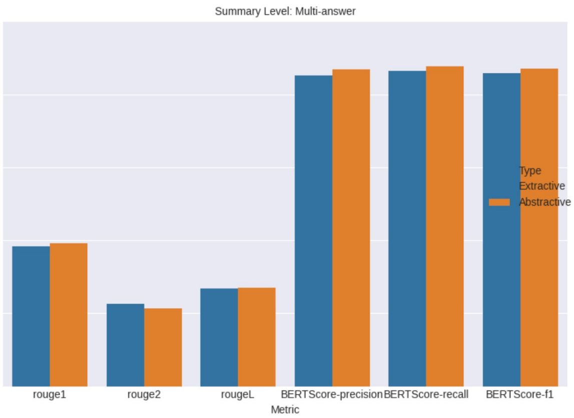

Figure 2: Metric comparison for Multi-answer summarization.

8 Discussion

Our results demonstrate several key findings.

First, extractive methods outperform abstractive ones on ROUGE due to their reliance on direct sentence selection, which is less sensitive to input truncation. However, the abstractive model achieves competitive BERTScore (6.1 Table 1), indicating strong semantic similarity even when surface-level overlap is low. This suggests that while extractive methods preserve exact phrasing, abstractive models can better capture overall meaning.

Second, multi-answer tasks exhibit performance degradation compared to single-answer tasks. This is primarily caused by the extreme length of combined inputs, which leads to significant information loss when truncated at the current 512-token limit. Multi-answer examples are truncated because of token limitation, directly harming extractive coverage and abstractive quality. Output length limits further exacerbate this problem, particularly for multi-answer summaries that require more space to convey all relevant information. Hierarchical summarization mitigates this issue by compressing individual answers before fusion, resulting in improved BERTScore and more coherent summaries.

Third, our analysis highlights several challenges for future work:

- **Input truncation:** Current token limits prevent models from accessing full context, which is critical in medical summarization tasks. Expanding input windows (1024–4096 tokens) would help preserve important details.
- **Resource constraints:** Limited compute memory restricts encoder length and output size, constraining both extractive and abstractive performance.
- **Long-context understanding:** Multi-answer summarization requires models capable of integrating dispersed information.

9 Conclusion

The evaluation shows that answer-level extractive summarization performs slightly better on ROUGE (ROUGE-1 0.52 vs. 0.49 for abstractive), while multi-answer performance drops noticeably (ROUGE-1 0.39 abstractive, 0.34 extractive). However, across all tasks, BERTScore remains consistently higher, with answer-level F1 scores around 0.89 (Figure 1) and multi-answer scores around 0.87 (Figure 2), demonstrating that the model preserves meaning even when surface overlap is lower. In other words, semantic quality is stronger than literal matching, highlighting better BERTScore performance compared to ROUGE.

These limitations are mainly due to the 512-token input cap caused by limited compute resources, which leads to severe truncation, especially in multi-answer examples. Increasing the token window (1024–4096) would retain more context and improve both accuracy and ROUGE overlap. Additionally, adopting a domain-specific long-context model like BioLongformer, trained on biomedical text, is likely to outperform the standard Longformer for medical summarization tasks.

AI Usage Statement

We used AI tools (ChatGPT) only for assistance with LaTeX formatting, grammar refinement, and improving the clarity of phrasing in the written report. AI tools were also used to research available summarization architectures and compare model families (e.g., Longformer, Bio- Longformer vs. T5) during the early design stage.

Computing Resources

Models were trained and evaluated using Google Colab GPUs. All reported experiments fit within these resource limits.

References

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